Assignment 3

Digital Signal processing

IIR filters

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This assignment covers IIR filters.

Your task is to solve a problem which requires real time filtering of a physical quantity. This analogue quantity is then digitised and filtered in realtime with IIR filters.

You have two options for the data acquisition:

1. Arduino. They work under Windows, Mac and Linux. They have a sampling rate of up to 1kHz. It measures analogue voltage.
2. Attys: Linux-only but have isolated inputs for biomedical measurements.

Every team (2 members) needs to have a different application for their measurement. Think of something simple such as measuring the speed of a fan with an LDR. Or if your flowerpot is watered. Add your topic to the wiki provided on moodle as soon as possible. It needs to solve a practical real life problem and requires low, high, bandpass or bandstop IIR filtering.

Again, you work in teams of two students and one report is submitted per team.

1. Present a real-time measurement problem to be solved which requires filtering. For example noise needs to be removed or a signal needs to be detected. Marks are given for initiative, inventiveness and originality. Document the experiment with (all compulsory):
   1. photos of the setup
   2. dataflow diagrams
   3. YouTube clip(s)

   Determine the filter response(s) which are required and justify them. Generate the SOS coefficients for the filter(s) Python's high level functions such as "butter" or do your own design math, for example, for a notch filter (see lecture). [25%]
2. Write a real time IIR filter class as a chain of 2nd order filters and add unit tests to the IIR class which test both their timedomain and frequency responses. Optimise this class for speed by testing different coding approaches and benchmark at least two approaches. [25%]
3. Add code which checks that your callback routine is faster than the sampling interval, warns the user if that's not the case while the program is running and explain why checking the sampling rate is essential for DSP. Optimise your entire processing pipeline. [25%]
4. Compare your filtered results with the original recordings, show both signals in a real time demo (YouTube clip) and discuss whether you have been successful. Do a critical analysis [25%]

High level *design* commands are allowed such as "butter". Any use of "lfilter", "conv" or other high

level python filter operation during the execution of the filter will result again in zero marks. Proof of real-time processing in the form of a video needs to be given and the video needs to show clearly what it's about. Please add your link to the wiki and also add a readme to your zip containing all the files.

If you use an Arduino then you must use pyFirmata2 as it provides event-driven real-time callbacks.

If you need an electrically isolated ADC for biomedical recordings then a limited number of Attys boards are available (only for Linux users).

The code needs to be again submitted as a zip file and it will be tested if it runs. Crashing code will result in low marks. As before we expect sharp figures in vector format in the report. The complete code needs to be in the appendix and also uploaded to moodle.

Make sure your code is:

- platform-independent
- does not contain absolute paths
- Code that crashes will result in low marks. The same applies to code which does not display any plots.
- No high level Python signal processing functions for the realtime processing in the callback are allowed. Any use of "lfilter" and any a-causal processing (i.e. data array in and data array out) commands will result in zero or very low marks. Do not use FFT- or FIR-filtering in this assignment. The purpose of this assignment is to assess your use of the IIR filter, so filtering must be carried out by designing and using an IIR filter. Any use of FFT- or FIR-filtering will result in zero marks for that part.
- The code needs to be efficient.
- Inflated/unreadable code generated by an LLM will result in low or zero marks. It is important that a human can understand the code generated. All tasks above can be solved elegantly with very few lines of code. If you use GenAI, then take its raw output and optimise it.
- GenAI tools may be used in this assignment. If you use GenAI, you must include a short acknowledgement in your report stating which tool(s) you used (name, version, publisher), how you used them, and what work you contributed yourself. See the UoG guidance on acknowledging GenAI use.
- We reserve the right to interview you to test if you own the code (i.e. that you can explain it).

Deadline for the report is Monday, 7th Dec, 9am.